



Math 111: Linear Algebra

First Exam, 31 October 2017

Duration: 90 minutes

- In questions 1-4, show all your work.
- In question 5, select the correct answer (no justification is required).
- Calculators and communication devices are not allowed in the examination room.

Name: _____

Student ID: _____

Section Number: _____ Serial Number: _____

Q1 (19 pts)	
Q2 (20 pts)	
Q3 (20 pts)	
Q4 (20 pts)	
Q5 (21 pts)	
Total	

1. [10 + 9 = 19 pts.] Let A and B be two $n \times n$ invertible matrices.

(a) Show that: $\text{adj}(AB) = \text{adj}(B)\text{adj}(A)$.

(b) Show that: if $(AB)^2 = A^2B^2$, then $AB = BA$.

2. [10 + 10 = 20 pts.] Let A be an $n \times n$ matrix.

(a) Show that: if A is invertible, then A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$.

(b) Show that: if the linear system $AX = B$ has exactly one solution for every $n \times 1$ matrix B , then A is invertible.

3. [10 + 10 = 20 pts.] Let $A = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 7 & 1 \end{bmatrix}$ and $C = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$.

(a) Show that $\det(2A) = 2 \begin{vmatrix} \text{tr}(A) & 1 \\ \text{tr}(A^2) & \text{tr}(A) \end{vmatrix}$.

(b) Find a 2×2 matrix X such that $A^T X - 3B = C$.

4. [10 + 10 = 20 pts.] Let $A = \begin{bmatrix} 1 & -1 & 2 \\ 1 & 0 & 1 \\ 2 & -3 & 5 \end{bmatrix}$.

(a) Find $\text{adj}(A)$.

(b) Solve the system $AX = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

5. [3 × 7 = 21 pts.] In each of the following questions, select the correct answer.

I. If $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 3 \\ 0 & 0 & 3 \end{bmatrix}$, then $\det(A)$ is:

(A) 0

(B) 3

(C) -3

(D) None of the above.

II. If A and B are $n \times n$ matrices with $\text{tr}(A) = 2$ and $\text{tr}(B) = 5$, then $\text{tr}(A + B)$ is:

(A) 3

(B) 7

(C) 10

(D) None of the above.

III. If A and B are 3×3 matrices with $\det(A) = 3$ and $\det(B) = 2$, then $\det(2A^T B^{-1})$ is:

- (A) 3
- (B) 6
- (C) 12
- (D) None of the above.

IV. The maximum number of zero entries that a 3×3 invertible matrix can have is:

- (A) 5
- (B) 7
- (C) 9
- (D) None of the above.

V. If A is an $n \times n$ matrix such that $A^T A = A$, then:

- (A) $A^2 = A$
- (B) $A = -A$
- (C) $A = I + 2A$
- (D) None of the above.

VI. If $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, then the total number of all-zero rows in the reduced row echelon form of A is:

- (A) 0
- (B) 1
- (C) 2
- (D) None of the above.

VII. The total number of inversions in the permutation 43125 of $S = \{1, 2, 3, 4, 5\}$ is:

- (A) 3
- (B) 4
- (C) 5
- (D) None of the above.

Solutions

1. [10 + 9 = 20 pts.] Let A and B be two $n \times n$ invertible matrices.

(a) Show that: $\text{adj}(AB) = \text{adj}(B)\text{adj}(A)$.

$$\text{adj}(B)\text{adj}(A) = |B|B^{-1}|A|A^{-1} = |AB|B^{-1}A^{-1} = |AB|(AB)^{-1} = \text{adj}(AB).$$

(b) Show that: if $(AB)^2 = A^2B^2$, then $AB = BA$.

If $(AB)^2 = (AB)(AB) = A^2B^2$, then multiplying both sides of this equation from the left by A^{-1} yields $B(AB) = AB^2$. Now multiplying both sides of the last equation from the right by B^{-1} yields $BA = AB$.

2. [10 + 5 + 5 = 20 pts.] Let A be an $n \times n$ matrix.

(a) Show that: if A is invertible then A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$.

If A is invertible, then A^{-1} exists, and $AA^{-1} = A^{-1}A = I$. Taking transposes we get: $(AA^{-1})^T = I^T$ and $(A^{-1}A)^T = I^T$. That is $(A^{-1})^T A^T = I$ and $A^T (A^{-1})^T = I$. Thus, A^T is invertible, and since the inverse of a matrix, if it exists, is unique we have $(A^T)^{-1} = (A^{-1})^T$.

(b) Show that: if the linear system $AX = B$ has exactly one solution for every $n \times 1$ matrix B , then A is invertible.

If the system $AX = B$ has exactly one solution for every $n \times 1$ matrix B , then in particular this is so for the systems

$$AX = \text{col}_1(I), AX = \text{col}_2(I), \dots, AX = \text{col}_n(I).$$

Let $X_1, X_2, \dots, X_n$ be solutions of the respective systems, and let us form an $n \times n$ matrix $C = [X_1 | X_2 | \dots | X_n]$. Then

$$AC = [AX_1 | AX_2 | \dots | AX_n] = I.$$

Thus, it follows that $C = A^{-1}$. Hence, A is invertible.

3. [10 + 10 = 20 pts.] Let $A = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 7 & 1 \end{bmatrix}$ and $C = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$.

(a) Show that $\det(2A) = 2 \begin{vmatrix} \text{tr}(A) & 1 \\ \text{tr}(A^2) & \text{tr}(A) \end{vmatrix}$.

We have $|A| = 1$, $\text{tr}(A) = 2$ and $\text{tr}(A^2) = \text{tr}\left(\begin{bmatrix} 3 & -2 \\ 2 & -1 \end{bmatrix}\right) = 2$.

Since $|2A| = 2^2|A| = 4$ and $2 \begin{vmatrix} \text{tr}(A) & 1 \\ \text{tr}(A^2) & \text{tr}(A) \end{vmatrix} = 2 \begin{vmatrix} 2 & 1 \\ 2 & 2 \end{vmatrix} = 2(2) = 4$, the equality holds.

(b) Find a 2×2 matrix X such that $A^T X - 3B = C$.

$A^T X - 3B = C \Rightarrow A^T X = 3B + C \Rightarrow X = (A^T)^{-1}(3B + C) = (A^{-1})^T(3B + C)$. Thus,

$$X = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}^T \left(3 \begin{bmatrix} 1 & 0 \\ 7 & 1 \end{bmatrix} + \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix} \right) = \begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 8 & 4 \\ 22 & 5 \end{bmatrix} = \begin{bmatrix} -22 & -5 \\ 52 & 14 \end{bmatrix}.$$

4. [10 + 10 = 20 pts.] Let $A = \begin{bmatrix} 1 & -1 & 2 \\ 1 & 0 & 1 \\ 2 & -3 & 5 \end{bmatrix}$.

(a) Find $\text{adj}(A)$.

$$\text{adj}(A) = \begin{bmatrix} 3 & -1 & -1 \\ -3 & 1 & 1 \\ -3 & 1 & 1 \end{bmatrix}.$$

(b) Solve the homogeneous linear system $AX = O$.

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 1 & 0 & 1 & 0 \\ 2 & -3 & 5 & 0 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]. \text{ Thus, the general solution of the given system}$$

is $X = \begin{bmatrix} -t \\ t \\ t \end{bmatrix}$, where $t \in \mathbb{R}$.

5. [$3 \times 7 = 21$ pts.] In each of the following questions, select the correct answer.

I. If $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 3 \\ 0 & 0 & 3 \end{bmatrix}$, then $\det(A)$ is:

(A) 0

(B) 3

(C) -3

(D) None of the above.

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(A) 3

(B) 7

(C) 10

(D) None of the above.

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(A) 3

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(C) 12

(D) None of the above.

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